In [3]:
```
import pandas as pd
import matplotlib.pylab as plt
import numpy as np
```

In [4]:
```
df = pd.read_csv("autodata.csv")
```

In [5]:
```
df.head(5)
```

Out[5]:

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	engine-location	whee bas
0	0	3	122	alfa-romero	std	two	convertible	rwd	front	88.
1	1	3	122	alfa-romero	std	two	convertible	rwd	front	88.
2	2	1	122	alfa-romero	std	two	hatchback	rwd	front	94.
3	3	2	164	audi	std	four	sedan	fwd	front	99.
4	4	2	164	audi	std	four	sedan	4wd	front	99.

5 rows × 30 columns

In [6]:
```
df.tail(5)
```

Out[6]:

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base
196	196	-1	95	volvo	std	four	sedan	rwd	front	109.1
197	197	-1	95	volvo	turbo	four	sedan	rwd	front	109.1
198	198	-1	95	volvo	std	four	sedan	rwd	front	109.1
199	199	-1	95	volvo	turbo	four	sedan	rwd	front	109.1
200	200	-1	95	volvo	turbo	four	sedan	rwd	front	109.1

5 rows × 30 columns

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 201 entries, 0 to 200
Data columns (total 30 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         201 non-null    int64
 1   symboling          201 non-null    int64
 2   normalized-losses  201 non-null    int64
 3   make               201 non-null    object
 4   aspiration         201 non-null    object
 5   num-of-doors       201 non-null    object
 6   body-style         201 non-null    object
 7   drive-wheels       201 non-null    object
 8   engine-location    201 non-null    object
 9   wheel-base         201 non-null    float64
 10  length             201 non-null    float64
 11  width              201 non-null    float64
 12  height             201 non-null    float64
 13  curb-weight        201 non-null    int64
 14  engine-type        201 non-null    object
 15  num-of-cylinders   201 non-null    object
 16  engine-size        201 non-null    int64
 17  fuel-system        201 non-null    object
 18  bore               201 non-null    float64
 19  stroke             197 non-null    float64
 20  compression-ratio  201 non-null    float64
 21  horsepower         199 non-null    float64
 22  peak-rpm           199 non-null    float64
 23  city-mpg           201 non-null    int64
 24  highway-mpg        201 non-null    int64
 25  price              201 non-null    float64
 26  city-L/100km       201 non-null    float64
 27  horsepower-binned  199 non-null    object
 28  diesel             201 non-null    int64
 29  gas                201 non-null    int64
dtypes: float64(11), int64(9), object(10)
memory usage: 47.2+ KB
```

In [8]:
```
df.describe()
```

Out[8]:

	Unnamed: 0	symboling	normalized-losses	wheel-base	length	width	height	cu
count	201.000000	201.000000	201.00000	201.000000	201.000000	201.000000	201.000000	20
mean	100.000000	0.840796	122.00000	98.797015	0.837102	0.915126	53.766667	255
std	58.167861	1.254802	31.99625	6.066366	0.059213	0.029187	2.447822	51
min	0.000000	-2.000000	65.00000	86.600000	0.678039	0.837500	47.800000	148
25%	50.000000	0.000000	101.00000	94.500000	0.801538	0.890278	52.000000	216
50%	100.000000	1.000000	122.00000	97.000000	0.832292	0.909722	54.100000	241
75%	150.000000	2.000000	137.00000	102.400000	0.881788	0.925000	55.500000	292
max	200.000000	3.000000	256.00000	120.900000	1.000000	1.000000	59.800000	406

In [9]:
```
df.isnull()
```

Out[9]:

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	.
0	False	False	False	False	False	False	False	False	False	False	.
1	False	False	False	False	False	False	False	False	False	False	.
2	False	False	False	False	False	False	False	False	False	False	.
3	False	False	False	False	False	False	False	False	False	False	.
4	False	False	False	False	False	False	False	False	False	False	.
...	...	...	...	...	...	...	...	...	...	...	.
196	False	False	False	False	False	False	False	False	False	False	.
197	False	False	False	False	False	False	False	False	False	False	.
198	False	False	False	False	False	False	False	False	False	False	.
199	False	False	False	False	False	False	False	False	False	False	.
200	False	False	False	False	False	False	False	False	False	False	.

201 rows × 30 columns

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Unnamed: 0            0
symboling             0
normalized-losses     0
make                  0
aspiration            0
num-of-doors          0
body-style            0
drive-wheels          0
engine-location       0
wheel-base            0
length                0
width                 0
height                0
curb-weight           0
engine-type           0
num-of-cylinders      0
engine-size           0
fuel-system           0
bore                  0
stroke                4
compression-ratio     0
horsepower            2
peak-rpm              2
city-mpg              0
highway-mpg           0
price                 0
city-L/100km          0
horsepower-binned     2
diesel                0
gas                   0
dtype: int64
```

In [11]:
```
avg_stroke = df["stroke"].astype("float").mean(axis = 0)
```

In [12]:
```
print("Average of stroke : ",avg_stroke)
```

```
Average of stroke :  3.256903553299492
```

In [13]:
```
avg_hp = df["horsepower"].astype("float").mean(axis = 0)
print("Average of stroke:", avg_hp)
```

```
Average of stroke: 103.39698492462311
```

In [14]:
```
df["horsepower"].replace(np.nan, avg_hp, inplace = True)
```

In [15]:
```
avg_rpm = df["peak-rpm"].astype("float").mean(axis = 0)
print("Average of stroke:", avg_rpm)
```

```
Average of stroke: 5117.587939698493
```

In [18]:
```
df["peak-rpm"].replace(np.nan,avg_hp.inplace="true")
```

```
  Cell In[18], line 1
    df["peak-rpm"].replace(np.nan,avg_hp.inplace="true")
                                         ^
SyntaxError: expression cannot contain assignment, perhaps you meant "=="?
```

In [19]:
```
df['num-of-doors'].value_counts()
```

Out[19]:
```
num-of-doors
four    115
two      86
Name: count, dtype: int64
```

In [20]:
```
df['num-of-doors'].value_counts().idxmax()
```

Out[20]:
```
'four'
```

In [21]:
```
df["num-of-doors"].replace(np.nan, "four", inplace=True)
df.dropna(subset=["horsepower-binned"], axis=0, inplace=True)
df.reset_index(drop=True, inplace=True)
```

In [22]:
```
df.isnull().sum()
```

Out[22]:

```
Unnamed: 0           0
symboling            0
normalized-losses    0
make                 0
aspiration           0
num-of-doors         0
body-style           0
drive-wheels         0
engine-location      0
wheel-base           0
length               0
width                0
height               0
curb-weight          0
engine-type          0
num-of-cylinders     0
engine-size          0
fuel-system          0
bore                 0
stroke               4
compression-ratio    0
horsepower           0
peak-rpm             0
city-mpg             0
highway-mpg          0
price                0
city-L/100km         0
horsepower-binned    0
diesel               0
gas                  0
dtype: int64
```

In [23]:

```
df['city-L/100km'] = 235/df["city-mpg"]
df.head()
```

Out[23]:

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	engine-location	whee bas
0	0	3	122	alfa-romero	std	two	convertible	rwd	front	88.
1	1	3	122	alfa-romero	std	two	convertible	rwd	front	88.
2	2	1	122	alfa-romero	std	two	hatchback	rwd	front	94.
3	3	2	164	audi	std	four	sedan	fwd	front	99.
4	4	2	164	audi	std	four	sedan	4wd	front	99.

5 rows × 30 columns

In [24]:

```
df['highway-L/100km'] = 235/df["highway-mpg"]
df.head()
```

Out[24]:

	Unnamed: 0	symboling	normalized-losses	make	aspiration	num-of-doors	body-style	drive-wheels	engine-location	whee bas
0	0	3	122	alfa-romero	std	two	convertible	rwd	front	88.
1	1	3	122	alfa-romero	std	two	convertible	rwd	front	88.
2	2	1	122	alfa-romero	std	two	hatchback	rwd	front	94.
3	3	2	164	audi	std	four	sedan	fwd	front	99.
4	4	2	164	audi	std	four	sedan	4wd	front	99.

5 rows × 31 columns

In [28]:
```
df['length']=df['length']/df['length'].max()
df['width']=df['width']/df['width'].max()
```

In [30]:
```
df['height']=df['height']/df['height'].max()
df[["length","width","height"]].head()
```

Out[30]:

	length	width	height
0	0.811148	0.890278	0.816054
1	0.811148	0.890278	0.816054
2	0.822681	0.909722	0.876254
3	0.848630	0.919444	0.908027
4	0.848630	0.922222	0.908027

In [31]:
```
df.columns
```

Out[31]:
```
Index(['Unnamed: 0', 'symboling', 'normalized-losses', 'make', 'aspiration',
       'num-of-doors', 'body-style', 'drive-wheels', 'engine-location',
       'wheel-base', 'length', 'width', 'height', 'curb-weight', 'engine-typ
e',
       'num-of-cylinders', 'engine-size', 'fuel-system', 'bore', 'stroke',
       'compression-ratio', 'horsepower', 'peak-rpm', 'city-mpg',
       'highway-mpg', 'price', 'city-L/100km', 'horsepower-binned', 'diese
l',
       'gas', 'highway-L/100km'],
      dtype='object')
```

In [34]:
```
df['aspiration'].value_counts()
```

Out[34]:
```
aspiration
std       163
turbo      36
Name: count, dtype: int64
```

In [35]:
```
dummy_variable_1=pd.get_dummies(df["aspiration"])
```

In [36]:
```
dummy_variable_1.head()
```

Out[36]:

	std	turbo
0	True	False
1	True	False
2	True	False
3	True	False
4	True	False

In [38]:
```
df=pd.concat([df,dummy_variable_1],axis=1)
```

In [39]:
```
df.drop("aspiration", axis = 1, inplace=True)
```

In [40]:
```
df.head()
```

Out[40]:

	Unnamed: 0	symboling	normalized-losses	make	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	lengtl
0	0	3	122	alfa-romero	two	convertible	rwd	front	88.6	0.811148
1	1	3	122	alfa-romero	two	convertible	rwd	front	88.6	0.811148
2	2	1	122	alfa-romero	two	hatchback	rwd	front	94.5	0.822681
3	3	2	164	audi	four	sedan	fwd	front	99.8	0.848630
4	4	2	164	audi	four	sedan	4wd	front	99.4	0.848630

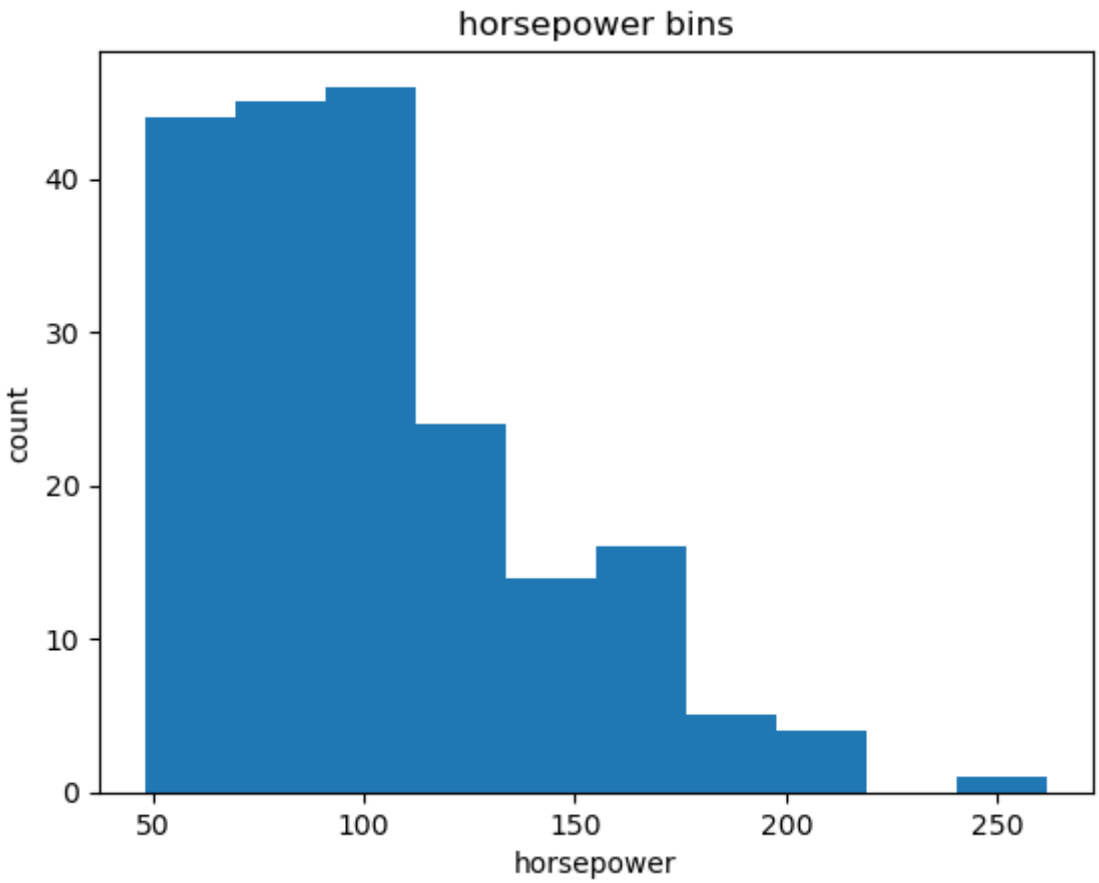
5 rows × 32 columns

In [44]:
```
df["horsepower"]=df["horsepower"].astype(float, copy=True)

%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
plt.pyplot.hist(df["horsepower"])

plt.pyplot.xlabel("horsepower")
plt.pyplot.ylabel("count")
plt.pyplot.title("horsepower bins")
```

Out[44]:
```
Text(0.5, 1.0, 'horsepower bins')
```



In [45]:
```
bins = np.linspace(min(df["horsepower"]), max(df["horsepower"]), 4)
bins
```

Out[45]:
```
array([ 48.        , 119.33333333, 190.66666667, 262.        ])
```

In [46]:
```
group_names = ['Low', 'Medium', 'High']
```

In [47]:
```
df['horsepower-binned'] = pd.cut(df['horsepower'], bins, labels=group_names,
df[['horsepower','horsepower-binned']].head(20)
```

Out[47]:

	horsepower	horsepower-binned
0	111.0	Low
1	111.0	Low
2	154.0	Medium
3	102.0	Low
4	115.0	Low
5	110.0	Low
6	110.0	Low
7	110.0	Low
8	140.0	Medium
9	101.0	Low
10	101.0	Low
11	121.0	Medium
12	121.0	Medium
13	121.0	Medium
14	182.0	Medium
15	182.0	Medium
16	182.0	Medium
17	48.0	Low
18	70.0	Low
19	70.0	Low

In [48]:

```
df["horsepower-binned"].value_counts()
```

Out[48]:

```
horsepower-binned
Low        151
Medium      43
High         5
Name: count, dtype: int64
```

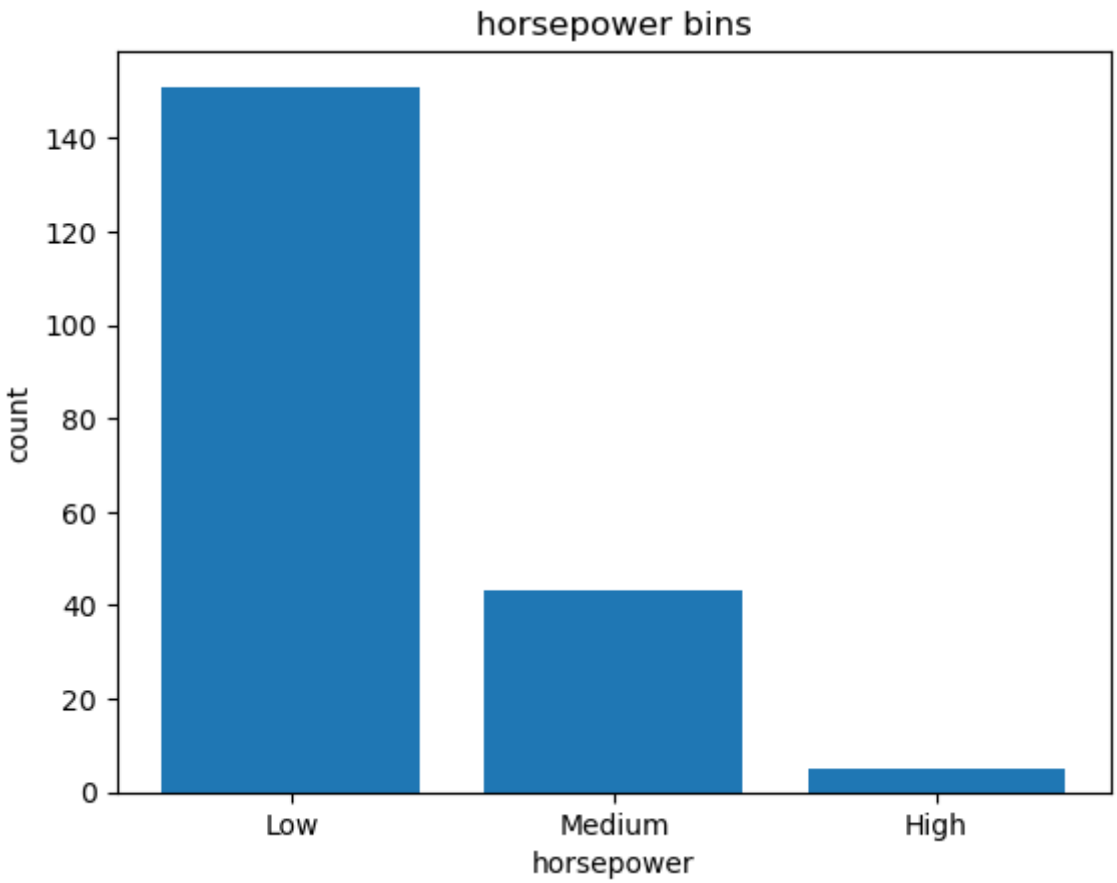
In [49]:

```
%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
pyplot.bar(group_names, df["horsepower-binned"].value_counts())

plt.pyplot.xlabel("horsepower")
plt.pyplot.ylabel("count")
plt.pyplot.title("horsepower bins")
```

Out[49]:

```
Text(0.5, 1.0, 'horsepower bins')
```

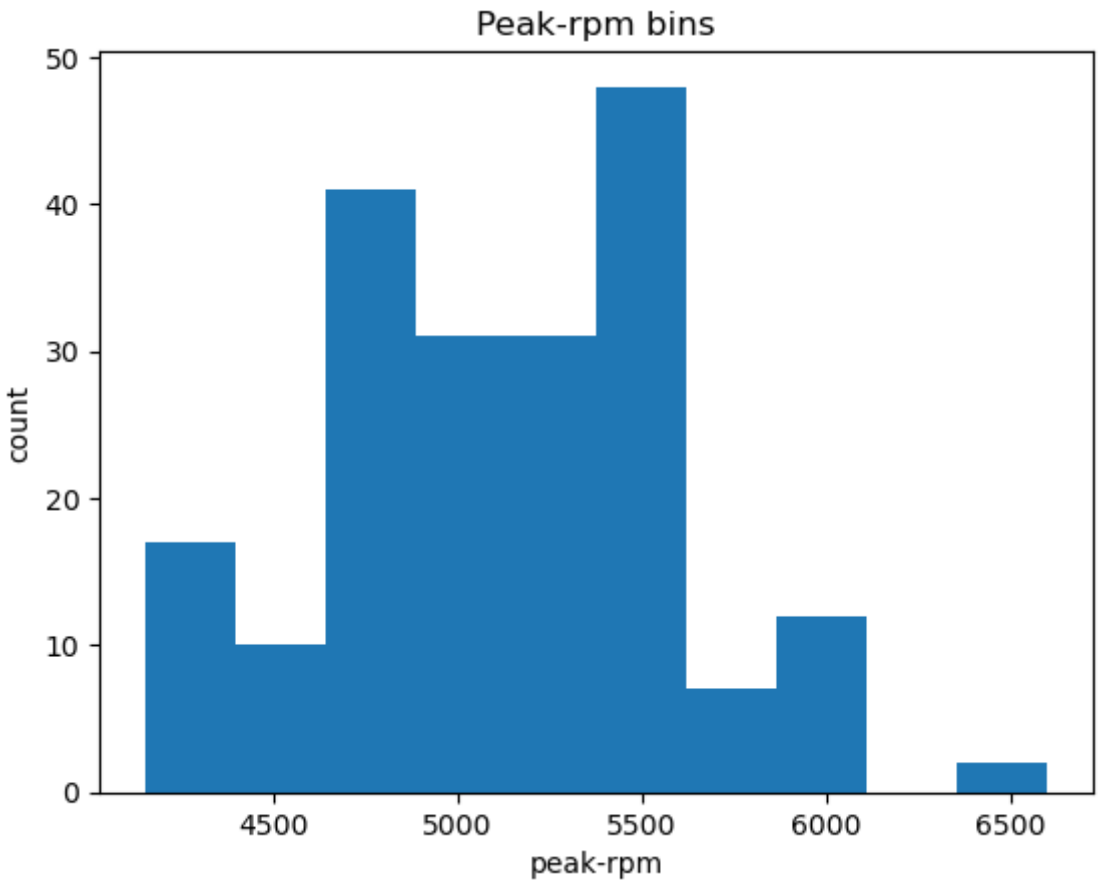


In [50]:
```
df["peak-rpm"]=df["peak-rpm"].astype(float, copy=True)
```

In [51]:
```
%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
plt.pyplot.hist(df["peak-rpm"])

plt.pyplot.xlabel("peak-rpm")
plt.pyplot.ylabel("count")
plt.pyplot.title("Peak-rpm bins")
```

Out[51]:
```
Text(0.5, 1.0, 'Peak-rpm bins')
```



In [52]:
```
bins = np.linspace(min(df["peak-rpm"]), max(df["peak-rpm"]), 4)
bins
```

Out[52]:
```
array([4150.        , 4966.66666667, 5783.33333333, 6600.        ])
```

In [53]:
```
group_names1 = ['Low', 'Medium', 'High']
```

In [54]:
```
df['peakrpm-binned'] = pd.cut(df['peak-rpm'], bins, labels=group_names, incl
df[['peak-rpm','peakrpm-binned']].head(20)
```

Out[54]:

	peak-rpm	peakrpm-binned
0	5000.0	Medium
1	5000.0	Medium
2	5000.0	Medium
3	5500.0	Medium
4	5500.0	Medium
5	5500.0	Medium
6	5500.0	Medium
7	5500.0	Medium
8	5500.0	Medium
9	5800.0	High
10	5800.0	High
11	4250.0	Low
12	4250.0	Low
13	4250.0	Low
14	5400.0	Medium
15	5400.0	Medium
16	5400.0	Medium
17	5100.0	Medium
18	5400.0	Medium
19	5400.0	Medium

In [55]:

```
df["peakrpm-binned"].value_counts()
```

Out[55]:

```
peakrpm-binned
Medium    109
Low        69
High       21
Name: count, dtype: int64
```

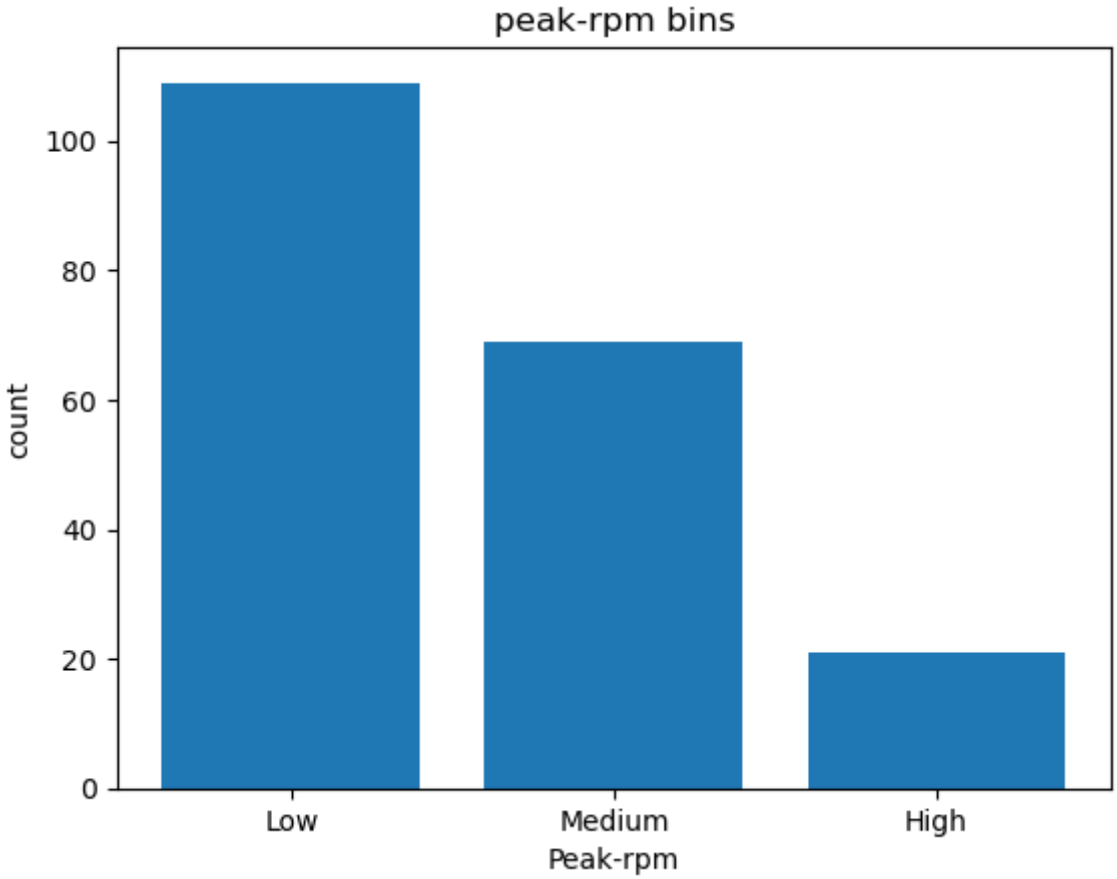
In [56]:

```
%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
pyplot.bar(group_names, df["peakrpm-binned"].value_counts())

plt.pyplot.xlabel("Peak-rpm")
plt.pyplot.ylabel("count")
plt.pyplot.title("peak-rpm bins")
```

Out[56]:

```
Text(0.5, 1.0, 'peak-rpm bins')
```

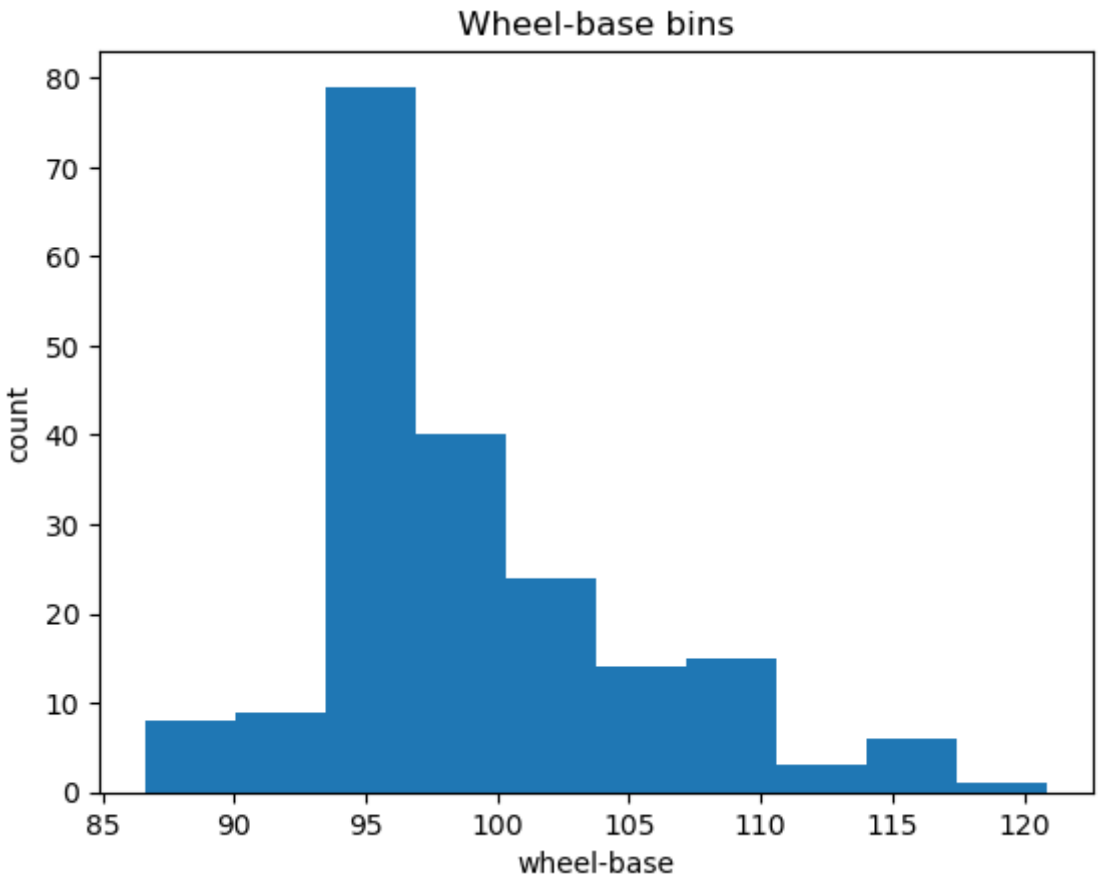


In [57]:
```
df["wheel-base"]=df["wheel-base"].astype(float, copy=True)
```

In [58]:
```
%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
plt.pyplot.hist(df["wheel-base"])

plt.pyplot.xlabel("wheel-base")
plt.pyplot.ylabel("count")
plt.pyplot.title("Wheel-base bins")
```

Out[58]:
```
Text(0.5, 1.0, 'Wheel-base bins')
```



In [59]:
```
bins = np.linspace(min(df["wheel-base"]), max(df["wheel-base"]), 4)
bins
```

Out[59]:
```
array([ 86.6        ,  98.03333333, 109.46666667, 120.9        ])
```

In [60]:
```
group_names = ['Low', 'Medium', 'High']
```

In [61]:
```
df['wheelbase-binned'] = pd.cut(df['wheel-base'], bins, labels=group_names,
df[['wheel-base','wheelbase-binned']].head(20)
```

Out[61]:

	wheel-base	wheelbase-binned
0	88.6	Low
1	88.6	Low
2	94.5	Low
3	99.8	Medium
4	99.4	Medium
5	99.8	Medium
6	105.8	Medium
7	105.8	Medium
8	105.8	Medium
9	101.2	Medium
10	101.2	Medium
11	101.2	Medium
12	101.2	Medium
13	103.5	Medium
14	103.5	Medium
15	103.5	Medium
16	110.0	High
17	88.4	Low
18	94.5	Low
19	94.5	Low

In [62]:

```
df["wheelbase-binned"].value_counts()
```

Out[62]:

```
wheelbase-binned
Low        114
Medium      72
High        13
Name: count, dtype: int64
```

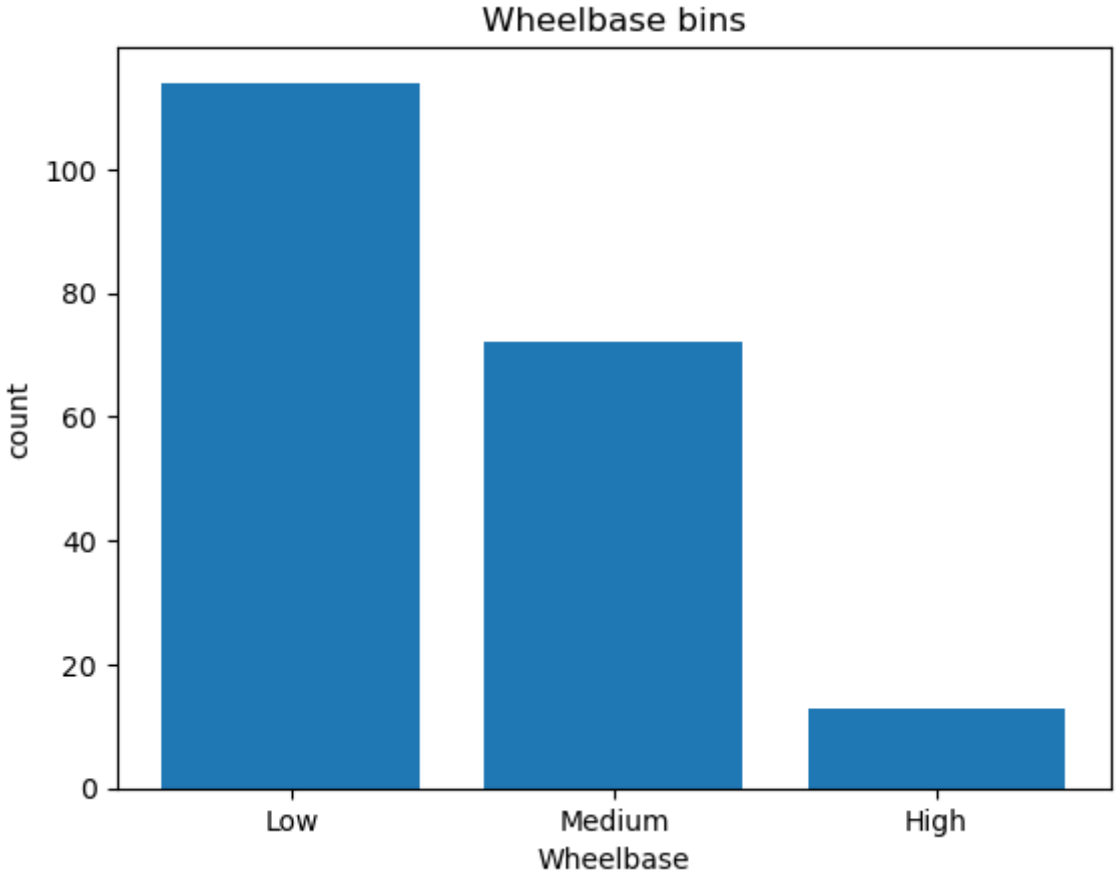
In [63]:

```
%matplotlib inline
import matplotlib as plt
from matplotlib import pyplot
pyplot.bar(group_names, df["wheelbase-binned"].value_counts())

plt.pyplot.xlabel("Wheelbase")
plt.pyplot.ylabel("count")
plt.pyplot.title("Wheelbase bins")
```

Out[63]:

```
Text(0.5, 1.0, 'Wheelbase bins')
```


Wheelbase bins
count
100
80
60
40
20
0
Low
Medium
High
Wheelbase

In [ ]: